

Towards logical association rule mining on ontology-based semantic trajectories

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Abstract—Mobility patterns have been investigated from multiple point-of-views, and different trajectory data mining paradigms have been proposed in the literature. Recent Semantic Trajectory representations rely on ontologies or RDF triples to represent different semantics, interlink data, and retrieve information. Nonetheless, current Trajectory Data Mining approaches use traditional transaction-based techniques that are not compatible with the relational nature of ontology-based representations. In this paper, we tackle the task of association rules mining by borrowing from the Knowledge Base Refinement field, the state-of-the-art AMIE 3 algorithm. After mining logical rules from a Foursquare dataset, we discuss the issues of applying off-the-shelf mining algorithms and discuss opportunities to develop a domain-tailored approach.

Index Terms—semantic trajectory; association rule mining; ontology;

I. INTRODUCTION

Many technologies generate or have been applied to generate data about the mobility of people, transportation vehicles, animals, and natural phenomena. Conceptual representations, tasks, algorithms, privacy issues, and data mining frameworks have been proposed and extensively investigated in the trajectory literature for the last two decades [1].

Nowadays, it has been long recognized by the trajectory data community that only spatiotemporal moving data, also called *raw trajectories*, are not enough to capture the complexity of humans' and other objects' behaviors. Contextual knowledge must be added in order to better understand trajectories [2] and to retrieve "business actionable knowledge" [3]. These *Semantic Trajectories* can be obtained by processing raw trajectories to extract abstract concepts, and possibly interlinking it with external data [1], [4].

In this work, we are interested in mining Association Rules from ontology-based Semantic Trajectories data. More specifically, we mine Horn rules, which are more expressive than conventional, transaction-based association rules.

Rule 1 is an example of a logical rule that we can mine. It states that given a visit to *The Orion Penthouse*¹ (a residential building in New York City) and a nearby place (within 2 kilometers) during the afternoon, this nearby place is usually (53% of the time) the *Times Square*². This rule incorporates different semantic dimensions (the venue's name and the visit's

time of the day) and uses a relation between the two visits (within 2 kilometers).

if	<i>visit V1 at The Orion Penthouse</i>	
and	<i>visit V2 during Afternoon</i>	
and	<i>V2 within 2 kilometers of V1</i>	(Rule 1)
then	<i>V2 at Times Square</i>	
	with 53% confidence	

To the best of our knowledge, mining association rules directly from the ontology-based trajectory representation has not been investigated by the trajectory community yet. Therefore, our first approach to mine rules explores the usage of AMIE 3 [5], a well-known state-of-the-art Knowledge Base rule mining algorithm.

II. BACKGROUND AND RELATED WORK

In this section, we review: (i) rule mining in Knowledge Bases; (ii) ontology-based modeling approaches for Semantic Trajectories; and (iii) works that mine or use rules in the Semantic Trajectory domain.

Knowledge base Rule Mining: Knowledge Bases (KBs) have been built by both private and public projects to serve multiple tasks [6]. Nonetheless, the data sources and data processing steps used to build them yield neither fully complete nor fully correct datasets. One of the approaches for Knowledge Base Refinement [6] is to mine association rules, which can describe general data regularities, check facts consistency, and infer new facts [7].

The authors of [5] divide the KB rule mining literature into two generations. The first one includes algorithms such as WARMR [8], which are based in Inductive Logic Programming (ILP). These algorithms have two problems when applied to current KBs: they do not scale to the millions of facts available, and they require negative facts as input, which are usually not stored in KBs. Furthermore, Knowledge Bases operate under the Open World Assumption (OWA), where a fact not present in the KB cannot be assumed as false [7].

These issues led to the development of the second generation of algorithms, starting with AMIE [7], which was specifically designed to work in the KB refinement context. The algorithm's implementation performance was improved by AMIE+ [9] and AMIE 3 [5].

¹<http://foursquare.com/v/the-orion-penthouse/4f93f1c8e5e828f50a2b81d1>

²<http://foursquare.com/v/times-square/49b7ed6df964a52030531fe3>

AMIE 3 implements runtime and memory optimizations, making it the state-of-the-art for mining logical rules from Resource Description Framework (RDF) triples. Nonetheless, it is essential to note that other approaches for mining rules from KB exist [10]–[12].

Ontology-based Semantic Trajectories: The first work to analyze a conceptual view of trajectories was [13]. It described semantic trajectories as a sequence of *stops* and *moves*, and used a relational schema for data storage. This further inspired works aiming to represent different semantic concepts or use alternative modeling techniques such as ontologies.

An ontology representation allows us to integrate and inter-link data from different sources [14], apply reasoning in order to infer new knowledge [4], and to adhere to the Semantic Web standards [15].

The FrameSTEP [4] is a framework for semantically enriching and querying trajectory data, supported by the STEP³ ontology. Different granularities can be expressed, and user-defined rules are applied to infer facts during queries. Complementary, the authors distribute the STEP ontology online, aligning it with the Semantic Web and Linked Open Data ideas.

Semantic trajectory rules: In the context of trajectory data, mining rules can be used as an automatic step to retrieve potentially interesting mobility patterns and moving objects' preferences. We consider that Semantic Trajectory Data Mining literature has explored rules in two contexts:

- *Handcrafted rules:* domain experts may use rules to capture domain knowledge. These rules can then be used, for example, to infer new facts [3], and to retrieve interesting entities [16]. These usages are associated with ontology-based representations since they offer languages for expressing complex rules.
- *Mining semantic trajectory rules:* many works mine rules by defining itemsets based on different trajectory semantic aspects and running the Apriori algorithm [17]–[20].

III. AMIE

AMIE is a multi-threaded, heavily memory-indexed algorithm to mine Horn Rules from an RDF triples dataset. We discuss the main properties of the original AMIE, which also apply to the AMIE 3 implementation. Based on [5], we present the following definitions:

Knowledge Base: A knowledge base $\mathcal{K}$ is a set of facts $r(s, o)$, where the subject s belongs to a set of entities $\mathcal{I}$, the relation r belongs to a set of relations $\mathcal{R}$, and the object o belongs to $\mathcal{I}$ or a set of literal values.

Relations and Functions: Given a relation r and the set of facts $r(s, o) \in \mathcal{K}$, the inverse relation of r , denoted r^- , consists of all the facts $r^-(o, s)$. A relation r is a function in $\mathcal{K}$ if r has at most one object for each subject. The notion of functions has been generalized to the *functionality* score [21] of a relation r :

$$fun(r) = \frac{|\{s : \exists o : r(s, o) \in \mathcal{K}\}|}{|\{(s, o) : r(s, o) \in \mathcal{K}\}|} \quad (\text{Eq. 1})$$

³<http://purl.org/net/step>

The functionality score is 1 for strict functions, close to 1 for quasi-functions, and it is smaller for relations that have many objects. If $fun(r) < fun(r^-)$, AMIE implicitly uses r^- during mining. By doing so, we can intuitively say that a fact $r(s, o)$ is a fact about s [7].

Atoms and Rules: An *atom* is an expression of the form $r(X, Y)$, where r is a relation and X, Y are either constants or variables. From now on, we denote variables by lowercase letters, whereas constants are always capitalized. An atom is *instantiated* if at least one of its arguments is a constant. A (conjunctive) *query* is a conjunction of atoms $\mathbf{B}$, which we separate by commas: $B_1, \dots, B_n$. A *substitution* σ , which can be applied to an atom or a query, is a partial mapping from variables to constants.

A (Horn) *rule* is a formula of the form $\mathbf{B} \Rightarrow H$, where $\mathbf{B}$ is a query of body atoms, and H is the head atom.

Predictions: Given a rule $R = B_1, \dots, B_n \Rightarrow H$, and a substitution σ , we call $\sigma(R)$ an *instantiation* of R . If $\sigma(B_i) \in \mathcal{K} \forall i \in \{1, \dots, n\}$, we call $\sigma(H)$ a *prediction* of R from $\mathcal{K}$, and write $\mathcal{K} \wedge R \models \sigma(H)$. If $\sigma(H) \in \mathcal{K}$, we call it a *true prediction*. If the prediction is a counter-example of the rule, we call it a *false prediction*.

Partial completeness assumption (PCA): The PCA was proposed in [7] to generate counter-examples when calculating rule's metrics. It considers that if the KB knows some r -attribute of s , then it knows all r -attributes of s . This is mostly true for relations with high functionality and a reasonable assumption for many relations mined by AMIE [9].

Support: The *support* of a rule R in a KB $\mathcal{K}$ is the number of true predictions p (of the form $r(X, Y)$) that the rule makes:

$$supp(R) = |\{p : (\mathcal{K} \wedge R \models p) \wedge p \in \mathcal{K}\}| \quad (\text{Eq. 2})$$

AMIE defines the proportional version of the absolute support, called *Head Coverage*. It is the ratio of instantiations of the head atom that are predicted by the rule:

$$hc(\mathbf{B} \Rightarrow r(x, y)) = \frac{supp(\mathbf{B} \Rightarrow r(x, y))}{|\{(x', y') : r(x', y') \in \mathcal{K}\}|} \quad (\text{Eq. 3})$$

Confidence: The confidence of a rule R in a KB $\mathcal{K}$ is the proportion of *true predictions* out of the *true predictions* and *false predictions*. AMIE uses two confidence metrics, Standard-Confidence (*Std Conf*) and PCA-Confidence (*PCA Conf*). They are given by Eq. 4, where cex denotes the counter-examples of R . In *Std Conf*, any predicted fact not in $\mathcal{K}$ is considered a counter-example. In the *PCA Conf*, only facts which contradicts the PCA are said to be counter-examples.

$$conf(R) = \frac{supp(R)}{supp(R) + |\{p : (\mathcal{K} \wedge R \models p) \wedge p \in cex(R)\}|} \quad (\text{Eq. 4})$$

AMIE has limited support for using the ontology schema. The mined rules may use the *rdf:type* relation to specify the domain and range of the head relation. Also, rules involving numerical expressions cannot be mined.

TABLE I
EXAMPLE OF CHECK-INS FROM ONE OF USER 13'S TRAJECTORIES.

Check-in	Venue	Venue category	Within Radius
13_1_1	ComputerLabEsLibrary ⁴	College/University	{13_1_2}
13_1_2	SpellmanHall ⁵	College/University	{13_1_1}

IV. PIPELINE

To mine and interpret logical rules from ontology-based semantic trajectories, we consider a 5-step data pipeline. The first step is composed of acquiring or constructing semantically-enriched trajectory data. This data is then represented and stored in an ontology-based representation, such as STEP, which is the focus of this work by its online availability and good documentation.

We propose a systematical approach to convert STEP to an application-specific representation, and then apply AMIE to mine rules. To make it viable to analyze the many mined patterns, we propose an approach to group similar rules based on metarules.

A. Semantically-enriched data

In our experiments, we use check-ins from the location-based social network Foursquare. A check-in is a digital spatiotemporal trail generated by a user interacting with a particular system, such as Foursquare or Twitter [14]. An example of semantic data is given in Table I.

We consider the main concepts of *User*, *Trajectory*, *Check-in*, and *Venue*. All of them can be associated with qualitative semantic descriptions, such as weather or check-in-related tags, as well as with each other. As discussed in section III, the limitations here are twofold: numerical data should be transformed into categorical data, and hierarchies cannot be mined. Therefore, we do not represent raw spatiotemporal data or venues' categories taxonomy.

B. Ontology-based representations

In STEP, a user (*Agent*) is associated with multiple trajectories, and each *Trajectory* can be associated with *Semantic Descriptions* through a *Feature Of Interest (FOI)* instance. Also, a trajectory needs to be associated with another *FOI* to connect with its *Episodes*. We use *ContextualElements* to represent *Venues*, and *Episodes* to represent *Check-ins*.

We extend the STEP ontology by adding subclasses of *FOI* for each of our usages. For example, to represent a venue's category, we create a *VenueCategory FOI* class. Finally, to connect two episodes, which in our case represent check-ins, we extend the *ContextualElement* with a *ContextualRelations* subclass. We represent the data from Table I in Figure 1a.

Note how the STEP representation requires many *intermediate* nodes. As discussed in [4], STEP uses inference

⁴<http://foursquare.com/v/computer-lab-es-library/4fa93241e4b02c573a9a87c9>

⁵<http://foursquare.com/v/spellman-hall/4d3a365dcc48224b400d3b4f>

rules to make it easier to query data. Nonetheless, mining Horn rules from this representation is very hard, since most semantic relationships are encoded as instances and classes, not relations.

It is important to note that other representations proposed in literature also have such issues since the schema, especially the relations, cannot be easily extended. Therefore, we propose a systematical approach to convert the STEP representation into an application-specific representation. The main goal of this process is to transform semantic instances (FOIs) into relations.

The result of applying this process in the ontology depicted in Figure 1a is presented in Figure 1b. We offer detailed pseudocode and implementation of this process on our website (see section V).

C. Mining rules

By analyzing initial experiments, we noted that not all entities should appear in a rule. More specifically, we consider uninteresting a rule which states a fact about a specific trajectory or check-in. In this context, we see the *Trajectory* and *Check-in* concepts as surrogate keys, defined to logically group related data. Therefore, we should not mine rules which contain instantiations of these concepts. As we are mainly interested in using an off-the-shelf algorithm, we do not make further modifications to AMIE.

D. Metarules

AMIE was originally used to mainly mine rules *without* constants. On the other hand, we are interested in rules which describe, for example, specific users, venues, or semantic aspects. Therefore, we mine multiple rules which follow the same pattern, but with different instantiated variables. For example, Rule 2 and Rule 3 express the same concept relations, but with different constants and atoms order.

$$r(a, b), q(b, VAL1) \Rightarrow p(VAL2, a) \quad (\text{Rule 2})$$

$$q(b, VAL3), r(a, b) \Rightarrow p(VAL4, a) \quad (\text{Rule 3})$$

Since AMIE's search-space strategy mines these rules independently, we propose a simple approach to group similar rules based on metarules. A metarule is a rule template which can be used to generate [22] or to group rules [23].

We say that a rule *R* *complies* with a metarule *M* if and only if it can be unified with *M* [22]. Here, we impose that *M* is a (Horn) rule with no instantiated atom. Given a set $\mathcal{R}$ of all mined rules, we can construct a set $\mathcal{M}$ of metarules such that every $R \in \mathcal{R}$ complies with one and only one $M \in \mathcal{M}$. For rules *R* without constants, *R* is its metarule.

Finally, we use $G(M)$ to refer to the subset of mined rules which complies with *M*. If $R \in G(M)$, we may represent *R* as a pair $\langle M, T \rangle$, where *T* is a tuple containing the instantiations in *R* accordingly to the order that they would appear in *M*. Considering Rule 2 and Rule 3, we can say that they are associated with the metarule $r(a, b), q(b, const0) \Rightarrow p(const1, a)$, and represented as $\langle VAL1, VAL2 \rangle$ and $\langle VAL3, VAL4 \rangle$.

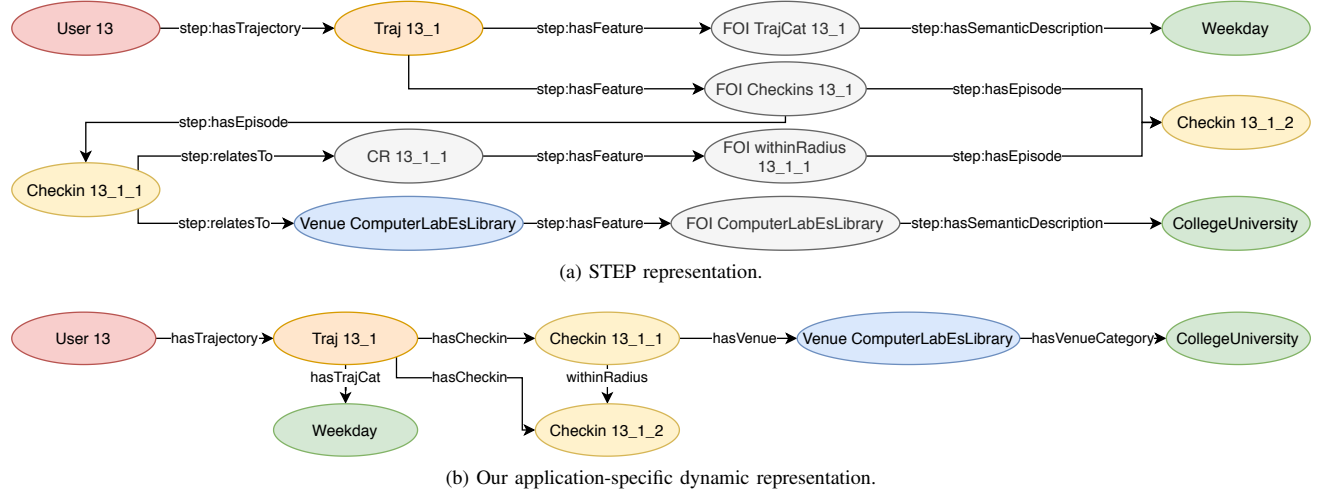


Fig. 1. Representation of Table I data. We omit the information related to the second check-in, including its *withinRadius* relation. Nodes with same colors in Figure 1a and Figure 1b represent similar semantics.

V. EXPERIMENTS

We begin by noting that all source code, datasets, and mined rules are available on our website⁶.

A. Setup

Dataset: we use the NYC Foursquare data collected by [24]. We filter the dataset by: removing duplicated check-ins (considering a 10-min threshold) [25]; removing venues with less than 5 check-ins; segmenting data into daily trajectories and selecting those with at least 5 check-ins; selecting those users' data with at least 10 trajectories.

The resulting data contains 158 *users*, with a total of 4,557 *trajectories*, and 35,460 *check-ins* at 4,378 different *venues*. Each of these concepts is represented by a unique identifier, and a venue's name is part of the venue's identifier. We represent the following semantic information:

- Each trajectory is associated with the *day of the week*, *month*, and *trajectory category*, which is a subset of {Weekday, Weekend, Holiday}.
- Each check-in has a *time of the day* [26], which is one of Morning (8:00-11:59), Afternoon (12:00-19:59), or Night (20:00-7:59).
- Each venue has one of Foursquare's root categories: Arts/Entertainment, College/University, Food, Nightlife Spot, Outdoors/Recreation, Professional/Other Places, Residence, Shop/Service, and Travel/Transport.

We investigate three different connections, calculated between the check-ins within the same trajectory, to represent relations that could not be mined with traditional rule mining. We connect check-ins that occur within a threshold distance using *withinRadius*, and check-ins that occur within a threshold time difference using *withinTimeWindow*.

We also establish an order between the check-ins' occurrence using *before*. We apply the thresholds of 2 kilometers

distance and 2 hours difference. Note that *withinRadius* and *withinTimeWindow* are symmetric, and *before* is transitive.

Pruning: Although AMIE proposes the relative support Head Coverage (HC), we use absolute support to prune rules. Our motivation is based on the following observation.

Since each trajectory has one or more check-ins, the number of check-ins grows faster than the number of trajectories. Given $R = \mathbf{B} \Rightarrow r(x, y)$, the HC normalizes the number of different true-predictions (x, y) by the number of different facts $r(x', y') \in \mathcal{K}$. Therefore, an HC threshold would be more likely, for example, to filter out rules about check-ins than those about trajectories.

We mine rules with at most 4 atoms, minimum support of 50, and minimum PCA Conf of 0.1 (AMIE's default value). Note that we choose the 4-atom limit due to performance constraints and that minimum support was empirically chosen but should be specified by a domain-expert in a real-case scenario.

RDF data: The populated ontology with all materialized facts is exported as RDF triples. Triples that state facts about the ontology schema are removed since they are not used.

B. Data representation and rule mining

Following our data pipeline, we represent the semantic trajectories using STEP, and then convert STEP data to our application-specific representation.

Considering the metrics in Table II, our application-specific representation, when compared to STEP: has many more relations, which were previously represented as FOIs in STEP; has fewer unique subjects and objects since we do not create intermediate nodes; and uses 220k triples less than STEP.

Finally, we apply AMIE 3 to mine rules with constants, but without instantiating Check-ins or Trajectories. AMIE mined a total of 459,256 rules in 10h22m⁷.

⁷We executed our experiment in a machine with Intel® Core™ i7, 32 GiB system memory, and 12 cores.

⁶<https://github.com/falcaopetri/st-larm>

TABLE II
METRICS FOR EACH DATA REPRESENTATION.

Representation	#Relations	#Triples	#Subjects	#Objects
STEP	5	747,011	266,799	266,674
Application-specific	11	524,765	44,553	44,428

C. Filtering and manual evaluation

We filter out 148,294 rules, which contain duplicated symmetric relations. For example, if AMIE mines a rule which contains the atom *withinRadius(a, b)*, it will also mine another rule which contains *withinRadius(b, a)*. We note that our filtering is not perfect and that the mining algorithm should be aware of symmetric relations.

Our metarule approach groups the 310,962 rules into 1,329 metarules. Even so, we consider it to be an unrealistic set to be analyzed. Manually checking some metarules shows that most of them state very generic facts. In special, we find that the set of relations *withinRadius*, *withinTimeWindow*, and *before* can be combined in multiple ways, and rapidly add many check-ins concepts. The result is multiple rules, which generically describes spurious correlations between many check-ins.

Following, we proceed by removing rules with the relation *before*, which is the choice that most reduces the number of metarules. After this process, we have 211,908 rules and 370 metarules to be analyzed. We classify each metarule into one of five qualitative categories, as discussed below.

Table III shows one metarule example for each category. Rule 4 associates a visit to a specific place at a specific time with one of Weekday, Weekend, or Holiday. Rule 5 associates visiting a specific venue with visiting another nearby place. Rule 6 shows a spurious correlation between visiting two venues at the same time of the day. Rule 7 associates different check-ins but providing no contextualization about their semantics. Rule 8 is a duplicated rule, equivalent to another mined rule which differs only by the *withinRadius(a, b)* atom. At total, we have:

- 34 *interesting* metarules that bring direct insights about a user's behavior, visits at some venue (Rule 4), or other general patterns that hold in the dataset;
- 117 *potentially interesting* metarules that indicate interesting patterns, but might also be a spurious correlation (Rule 5).
- 71 *uninteresting* metarules that are spurious correlation (Rule 6) or that describe the ontology or our semantic data (such as relating Tuesdays to Weekdays).
- 131 *too-general* metarules that could be refined to interesting rules if more atoms were allowed, or not mined if a better search-space traversal was used (Rule 7).
- 17 *should-not-happen* metarules that should not be mined. This includes rules where two different variables always have the same bindings and duplicated symmetric rules, which were not properly filtered (Rule 8). By default, AMIE does not enforce that different variables should be mapped to different constants [5].

VI. DISCUSSION

Although we have observed that using a general-purpose Horn rules mining algorithm yields interesting results, we consider essential the proposal and development of a domain-tailored algorithm. Next, we discuss our main findings in order to guide future algorithms design.

Semantic representation: Horn rules mining can be applied only if we have meaningful relations between concepts. Therefore, a custom algorithm should *explicitly* or *implicitly* consider a relation-centric data representation, possibly based on STEP or other available ontology.

Domain knowledge: Carefully analyzing the main concepts such as Moving Object, Trajectory, Episode, and Geographical Entities (such as a venue) should allow an improved search-space mining strategy, reduce the number of too general rules, and allow mining longer rules.

Schema information and numeric rules: Most algorithms developed after AMIE tries to better use ontology schema information, which might be useful, for example, to mine rules which consider a venue's category taxonomy. Another interesting feature is being able to mine numeric rules [27], which, for example, could learn a distance threshold for *withinRadius* which maximizes the rule's confidence metric.

Metrics: Data incompleteness and incompleteness-aware metrics should be carefully explored in the context of semantic trajectories [5]. Also, we note that the data representation is directly related to how we must interpret a rule. Consider the rule *hasCheckin(t, c), hasVenue(c, V) ⇒ hasTrajectory(U, t)*, which states that trajectories which contain a check-in to a specific venue *V*, belong to the user *U*. Note that it states a fact regarding *the number of trajectories with visits to this venue*, without considering *the number of visits to this venue*.

Post-processing: Our proposed metarule grouping, as well as other post-processing and visualization steps, should be investigated. For example, the metarule definition can be relaxed to use subsumption and build a lattice of related rules.

VII. CONCLUSION

This paper has discussed the opportunities of applying logical rule mining algorithms on ontology-based semantic trajectory data. Our proposed data pipeline mined some interesting patterns in an experiment using Foursquare data. Nonetheless, based on the large number of rules that state too general facts, we argue in favor of developing a domain-specific mining algorithm. This algorithm should take advantage of the KB refinement community's vast advancements and strategies while also considering specific trajectory domain knowledge.

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TABLE III

EXAMPLE OF METARULES AND RULES. WE RANDOMLY SELECTED ONE METARULE FOR EACH CATEGORY, AND TWO RULES FOR EACH METARULE. ABBREVIATIONS USED: I = INTERESTING, PI = POTENTIALLY INTERESTING, U = UNINTERESTING, T = TOO GENERAL, S = SHOULD NOT HAPPEN.

ID	Metarule	Label	Examples of rules substitutions	Head Coverage	Std Conf	Absolute Support	Body Support
Rule 4	$hasCheckin(t, c), hasVenue(c, const0),$ $hasTime(c, const1) \Rightarrow hasTrajCat(t, const2)$	I	$\langle SMB^1, Morning, Weekday \rangle$ $\langle NYPS^2, Afternoon, Weekday \rangle$	0.0158 0.0126	1.00 0.88	74 59	74 67
Rule 5	$hasCheckin(t, a), hasCheckin(t, b),$ $hasVenue(a, const0) \Rightarrow withinRadius(a, b)$	PI	$\langle Duke\ Ellington\ Blvd^3 \rangle$ $\langle SLC^4 \rangle$	0.0006 0.0006	0.86 0.69	77 79	90 115
Rule 6	$hasVenue(a, const0), hasVenue(b, const1),$ $hasTime(a, t) \Rightarrow hasTime(b, t)$	U	$\langle FDU^5, PABT^6 \rangle$ $\langle Cypress\ Hills^7, SUS^8 \rangle$	0.0034 0.0022	0.33 0.33	119 79	357 237
Rule 7	$hasVenue(a, const0), withinRadius(a, b),$ $withinTime(c, b) \Rightarrow withinTime(c, a)$	T	$\langle Yankeemanina \rangle$ $\langle United\ Cigars\ Inc^9 \rangle$	0.0006 0.0030	0.64 0.54	68 346	107 647
Rule 8	$hasTrajectory(const0, t), hasCheckin(t, b),$ $withinRadius(a, b) \Rightarrow hasTime(a, const1)$	S	$\langle User\ 951, Afternoon \rangle$ $\langle User\ 880, Night \rangle$	0.0029 0.0025	0.36 0.36	103 89	287 251

¹SMB = Subway - Manhattan Bridge

²NYPS = New York Penn Station

³Duke Ellington Blvd

⁴SLC = St. Luke's Roman Catholic Church

⁵FDU = Fairleigh Dickinson University

⁶PABT = Port Authority Bus Terminal

⁷Cypress Hills

⁸SUS = Subway - 14Th St/Union Square

⁹United Cigars Inc

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